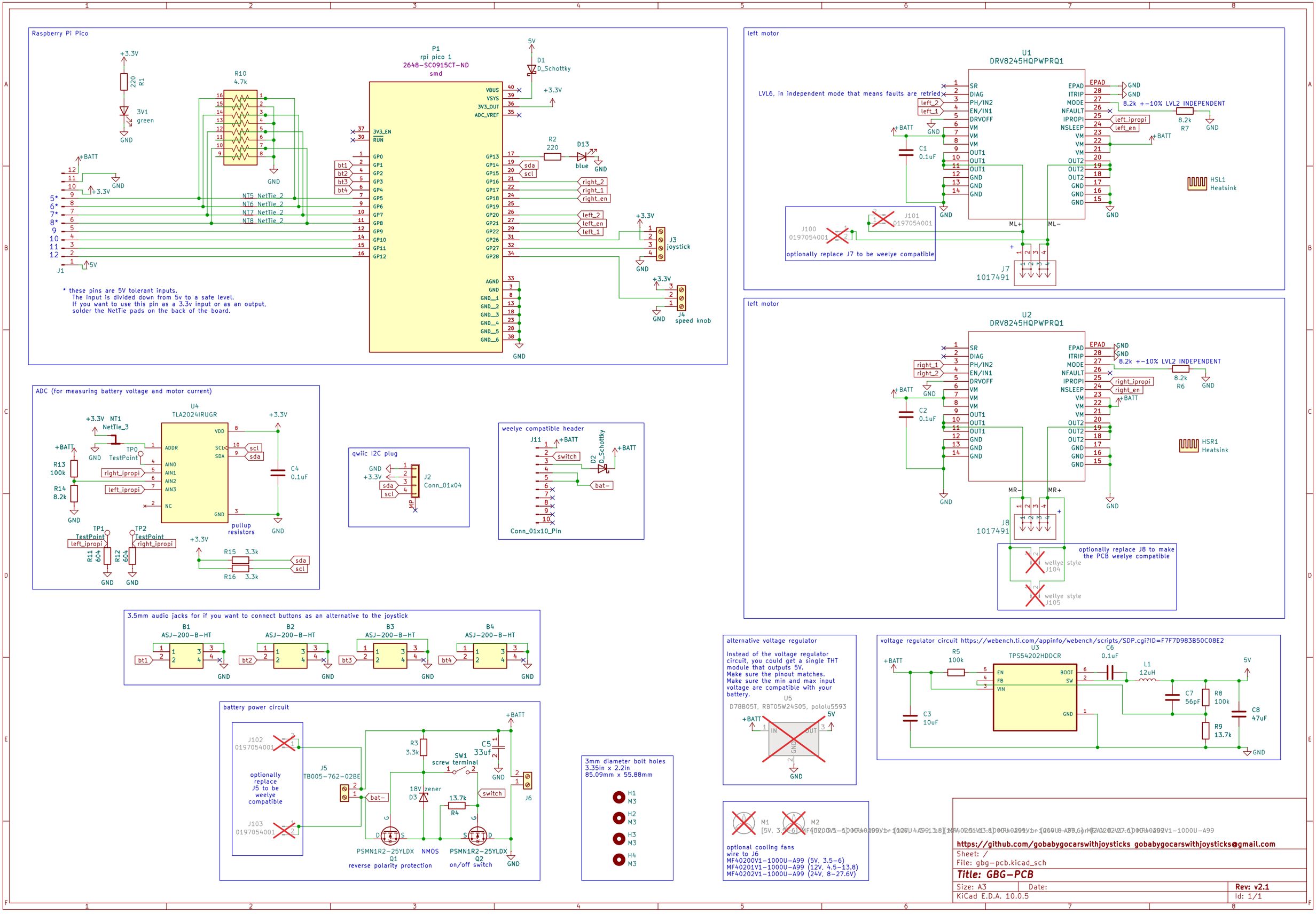

Raspberry Pi Pico
P1
rpi pico 1
2648-SC0915CT-ND
smd
R10
4.7k
R1
220
3V1
green
D1
D_Schottky
R2
220
D13
blue
J1
NT5 NetTie_2
NT6 NetTie_2
NT7 NetTie_2
NT8 NetTie_2
J3
joystick
J4
speed knob
* these pins are 5V tolerant inputs.
The input is divided down from 5v to a safe level.
If you want to use this pin as a 3.3v input or as an output,
solder the NetTie pads on the back of the board.
left motor
U1
DRV8245HQPWPRQ1
LVL6, in independent mode that means faults are retried
8.2k +-10% LVL2 INDEPENDENT
R7
C1
0.1uF
J100
0197054001
J101
0197054001
optionally replace J7 to be weelye compatible
J7
1017491
HSL1
Heatsink
left motor
U2
DRV8245HQPWPRQ1
8.2k +-10% LVL2 INDEPENDENT
R6
C2
0.1uF
J8
1017491
HSR1
Heatsink
optionally replace J8 to make
the PCB weelye compatible
wellye style
J104
wellye style
J105
ADC (for measuring battery voltage and motor current)
U4
TLA2024IRUGR
NT1
NetTie_3
TP0
TestPoint
R13
100k
R14
8.2k
C4
0.1uF
pullup
resistors
TP1
TestPoint
TP2
TestPoint
R11
604
R12
604
R15
3.3k
R16
3.3k
qwiic I2C plug
J2
Conn_01x04
weelye compatible header
J11
D2
D_Schottky
Conn_01x10_Pin
3.5mm audio jacks for if you want to connect buttons as an alternative to the joystick
B1
ASJ-200-B-HT
B2
ASJ-200-B-HT
B3
ASJ-200-B-HT
B4
ASJ-200-B-HT
battery power circuit
J102
0197054001
J103
0197054001
optionally
replace
J5 to be
weelye
compatible
J5
TB005-762-02BE
R3
3.3k
C5
33uf
SW1
screw terminal
18V zener
D3
13.7k
R4
J6
PSMN1R2-25YLDX
Q1
reverse polarity protection
NMOS
PSMN1R2-25YLDX
Q2
on/off switch
3mm diameter bolt holes
3.35in x 2.2in
85.09mm x 55.88mm
H1
M3
H2
M3
H3
M3
H4
M3
alternative voltage regulator
Instead of the voltage regulator
circuit, you could get a single THT
module that outputs 5V.
Make sure the pinout matches.
Make sure the min and max input
voltage are compatible with your
battery.
U5
D78B05T, RBT05W24S05, pololu5593
voltage regulator circuit https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=F7F7D983B50C08E2
U3
TPS54202HDDCR
R5
100k
C6
0.1uF
L1
12uH
C7
56pF
R8
100k
R9
13.7k
C8
47uF
C3
10uF
M1
M2
optional cooling fans
wire to J6
MF40200V1-1000U-A99 (5V, 3.5-6)
MF40201V1-1000U-A99 (12V, 4.5-13.8)
MF40202V1-1000U-A99 (24V, 8-27.6V)
https://github.com/gobabygocarswithjoysticks gobabygocarswithjoysticks@gmail.com
Sheet: /
File: gbg-pcb.kicad_sch
Title: GBG-PCB
Size: A3
Date:
Rev: v2.1
KiCad E.D.A. 10.0.5
Id: 1/1